

RBE 511 Final Project: MaxCal-Derived Swarm Control: Emergent Oscillation Between Coverage and Information Diffusion

Group 3

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Abstract

This project extends recent research on a theoretical decentralized framework called Maximum Caliber (MaxCal), which is used to design swarm behaviors from high-level goals while robots execute locally sampled transition rules derived from global constraints. Our approach derives the local control rule for each robot directly from the principle of MaxCal, which generalises the maximum-entropy principle from static state representations to trajectories and sequential processes. By imposing two macroscopic constraints and observing the coverage age on every cell of a discretised arena and an information age on every robot, we can optimize the MaxCal equation and obtain a stochastic transition kernel that the robots execute locally resulting in an oscillating behavior between the two tasks. With only the coverage constraint active, we analytically recover the predicted coverage of the arena and therefore the expected mixing rate of the underlying Markov chain; we additionally solve the inverse MaxCal problem for a uniform visitation target and recover the per-cell multiplier fields. For the information-diffusion layer, we solve the self-consistent fixed-point equation for the stationary distribution $\bar{\pi}$ and implement an inverse solver that returns the scalar λ_i that achieves a target information spreading rate. We aim to visually show that when two opposing behaviors, such as clustering and spreading, create a natural tradeoff, MaxCal can derive local behavior rules that achieve an optimal balance between them.

1 Introduction

Context. Decentralised swarm controllers are attractive whenever the operating environment is too large, too dynamic, or the ability to communicate is too lackluster to support centralised planning. Their design has taken

inspiration from the foraging-and-recruitment dynamics of social insects, in which individuals alternate between exploration and aggregation without explicit coordination [3, 4]. Within the complexity of swarm behaviors, area coverage and information diffusion have been studied independently, but no study has treated them within a single framework.

Problem. Existing controllers that address one objective or constraint typically do so at the expense of the other, and those that address both generally introduce hand-tuned switching rules or global synchronisation mechanisms that undermine the decentralised character of the swarm. This is particularly problematic with antagonistic behaviors, as is the case in coverage and clustering. Many approaches resolve this through the introduction of a global clock or a designated coordinator, which sacrifices the decentralized nature that makes swarms attractive in many problem cases, where each robot only has its local information.

Importance. The design of a decentralized controller derived from a set of constraints can allow us to decipher unknown optimal behaviors or work backwards into properly defining desired behaviors in a formalized manner. A controller whose behavior emerges from the imposed constraints is both theoretically cleaner and operationally more robust.

Difficulty. There are several challenges to this task, the first being that for the approach to work, the different constraints have to be reformulated into an incrementally additive form in order to use it with the MaxCal derivation. This can pose problems when the constraints are of different types. In our example of the cluster and the coverage, the age of information relates to the individual robots while the coverage age pertains to an individual cell. Second, the relaxation timescales of the two constraints depend on swarm size, robot speed, and

environment topology in different ways, so the existence of a stable oscillation is not guaranteed for arbitrary parameter combinations. Finally, a fully decentralised controller would require each robot to estimate global observables from only locally perceived information; we use globally computed multiplier fields for validation against theoretical predictions and recommend local estimation for future work.

Contribution. To show that a desired behavior can be obtained through constraining the MaxCal derivation, we derive a MaxCal-based hierarchical controller that imposes coverage and information diffusion constraints simultaneously. Each robot samples its Markov transitions locally from transition kernels obtained from the multipliers that are computed using the global target stationary distributions. We validate the controller in three stages: a Layer 1 stage that analyses both forward and inverse MaxCal theory on coverage and information-diffusion individually against analytical predictions; a linear Layer 2 stage that corresponds to the time-homogeneous baseline analysed by Pinciroli; and a full quadratic Layer 2 stage that produces the oscillation between coverage and clustering predicted by the theory [5]. We further compare linear and quadratic Layer 2 controllers quantitatively and identify the boundary conditions under which the oscillation emerges, breaks down into permanent spreading, or collapses into permanent clustering.

2 Related Work

This paper connects four areas of research: Maximum Caliber theory and its connection to Markov processes, entropy-based methods for deriving swarm rules, multi-robot coverage and information sharing, and decentralised swarm coordination. We review each area and explain how our work differs from existing results.

Maximum Caliber: core theory and Markov-chain derivations. The MaxCal principle extends Maximum Entropy (MaxEnt) from static distributions to distributions over entire trajectories. Pressé et al. [1] demonstrated that MaxCal recovers a broad set of results in non-equilibrium statistical physics from a single variational principle over path distributions, and

serves as the theoretical starting point for every constraint-derivation step in our controller. Ge et al. [2] prove the result most directly relevant to our implementation: when constraints are expressed as pairwise sequential statistics along trajectories, the process that maximises the caliber is always a Markov chain. This means the transition rule we derive is a valid Markov chain by construction, which is what allows us to apply standard convergence tools in Stage I. The broader applicability of MaxCal is surveyed by Dixit et al. [3], who apply it to gene networks, protein folding, and neural dynamics, and show that it works well as an inference tool when only partial observations of a system are available, which matches our scenario where robots have access to only local information. Ghosh et al. [4] complement this with a review that identifies when the Lagrange multipliers of the caliber optimization map to physically meaningful rate constants; we use their calibration approach to estimate the age-reduction rates r_C and r_I from short isolation runs.

Entropy, variational principles, and swarm design. Pinciroli [5] translates MaxCal into a practical swarm-robotics design tool: a designer writes down the global behaviour they want as a set of trajectory constraints, and the optimal local transition rule follows from the optimization. Our paper builds directly on this theory, and aims to test Pinciroli's predictions. A key finding of Pinciroli is that a time-homogeneous MaxCal solution always produces a fixed behavioural compromise and can never oscillate between modes, which motivates the hierarchical, quadratic-constraint architecture we implement and test. At the broader level, Bonabeau, Dorigo, and Theraulaz [6] show that the emergence of collective spatial structure in robot groups can be described as a phase transition, characterised by a Boltzmann-entropy index that tracks how organised the group is. This framing supports our use of the mean coverage age $\bar{k}_t$ and the Age of Information k_t as measurable quantities that separate the spreading phase from the clustering phase in our controller.

Multi-robot coverage. Cortés et al. [7] establish the geometric optimality criterion for multi-robot coverage, proving that gradient-descent controllers converge to

centroidal Voronoi configurations, where each robot settles at the weighted centre of its assigned region. This is a clean theoretical result, but it requires robots to know each other's positions and assumes continuous-time dynamics, neither of which holds in our setting. Hsieh et al. [8] address a more closely related problem consisting in redistributing a swarm across a discrete set of sites to match a target occupancy distribution using purely local, probabilistic transitions. Their Markov-chain controller shares the same goal as our Layer 1-C, but it is designed by hand rather than derived from a variational principle. Because we derive our kernel from MaxCal, we can extend it to include information diffusion constraints within the same framework, whereas Hsieh et al. offer no such mechanism. Neither paper addresses the tension between coverage and information freshness, nor produces a sustained alternation between spreading and clustering.

Age of Information and information diffusion.

Kaul, Yates, and Gruteser [9] introduce the Age of Information (AoI) metric, which measures how old the most recently received update is for a given receiver. Unlike delay or packet loss, AoI captures whether information is still relevant. We use AoI to track information freshness in the swarm, defining τ_t as the time since the last

pairwise robot encounter. The difference from Kaul et al. is that they optimize a transmission schedule over a fixed channel, whereas we move robots through space so that encounters, and thus AoI resets, emerge naturally from clustering. This means AoI and coverage are not independent and coupled in our system, with the Layer 2 mode-selection rule managing the trade-off between them. The mean-field encounter-rate geometric reformulation of Pinciroli [5] converts the swarm-level diffusion constraint into a condition on the probability that two robots share a cell at the same time, which makes the information diffusion controller locally computable without any direct communication between robots.

Markov chains and convergence. The mathematical tools we use in Stage I, mixing times, spectral gaps, and the convergence of empirical cell-visit frequencies, are drawn from Levin and Peres [10]. A key result from their framework is that the stationary distribution of a

random walk on an undirected graph is proportional to node degree. In our eight-connected grid, boundary and corner cells have fewer neighbours than interior cells, so robots naturally visit interior cells more often at steady state. This gives us a concrete, testable prediction for the stationary distribution of a random walk: $\overline{\pi}_k \propto \deg(k)$, which varies across the grid and is therefore easier to falsify than a uniform target would be. We use this as the ground truth against which our Stage I simulation results for area coverage are compared.

Positioning our contribution. The MaxCal literature has developed the mathematical foundations for path-entropy optimization but has not applied them to the combined coverage-and-diffusion problem in robotics. The swarm robotics literature has addressed coverage and information diffusion as separate problems, without a single variational framework that handles both. The AoI literature has formalised information freshness rigorously but always in the context of fixed communication networks, not mobile robots whose encounters must be engineered through motion. Our work closes all three gaps in a single analytically derived decentralised controller, and provides the first empirical map of the oscillatory parameter regime predicted by Pinciroli [5].

3 Methodology

3.1 Problem Formulation

We define the workspace as divided into K regions, and the state s_t of an individual robot is the index of the region it occupies at time t . A swarm of homogeneous robots evolves on this region graph, and each robot's local rule is obtained by applying the MaxCal principle to a chosen set of macroscopic constraints on the trajectory. Let $f_i(\gamma)$ denote an observable of a state trajectory $\gamma = \{s_0, s_1, \dots, s_N\}$ with target value F_i . Maximising the caliber $C(\Gamma)$ subject

$$\langle f_i(\gamma) \rangle = F_i \forall_i \text{ and } \sum_{\gamma \in \Gamma} p(\gamma) = 1 \text{ yields the}$$

optimal one-step transition rule

$$p^*(s_{t+1}|s_t) = \frac{1}{Z(s_t)} \exp[-\sum_i \lambda_i \Delta f_i(s_t, s_{t+1})],$$

where $Z(s_t)$ is a state transition function used as a normalizing factor over the next states s_{t+1} , and the λ_i are the Lagrange multipliers dual to the constraints. We adopt the hierarchical architecture proposed by Pinciroli [5], in which Coverage (C) and Information Diffusion (I) are Layer 1 behaviours and a Layer 2 controller alternates between them.

Layer 1 — Coverage. With multipliers λ_C^k coupled to the cumulative coverage observable and target stationary distribution, the MaxCal transition rule is

$$p^*(k_2|k_1) = \frac{w_{k_2}(k_1) \exp(-\lambda_C^{k_2})}{\sum_{k' \in N(k_1)} w_{k'}(k_1) \exp(-\lambda_C^{k'})},$$

where $N(k_1)$ is the local neighbourhood of k_1 and $w_{k_2}(k_1)$ is the connectivity weight, taken as unity throughout. Under the symmetric multiplier assignment $\lambda_C^k = \lambda_C \forall k$, this rule reduces to a uniform random walk whose stationary distribution is proportional to cell degrees, $\bar{\pi}_k \propto \deg(k)$. On a regular graph this gives $\bar{\pi}_k = 1/K$, and the empirical coverage

fraction $c_k(t) = \frac{1}{t} \sum_{t'=0}^{t-1} 1_k(s_{t'})$ satisfies the self-balancing relation

$$E[c_k(t+1) - c_k(t)] \approx \frac{1}{t+1} (\bar{\pi}_k - c_k(t)),$$

establishing convergence at rate $1/(t+1)$ in Markov time. When the target stationary distribution is uniform but the graph is irregular, the correct multiplier field is not $\lambda_C^k = 0$ but rather the solution of the inverse MaxCal stationary problem, which we describe and solve numerically in Sec. 3.2.

Layer 1 — Information Diffusion. After the mean-field encounter-rate reformulation of Pinciroli [5], the clustering behaviour takes the form

$$p^*(k_1|k_2) \propto \exp(-\lambda_C - \lambda_E \Delta f_E(k_1, k_2) - \lambda_I \bar{\pi}_{k_2})$$

where $\bar{\pi}_k$ acts as a pheromone-like visitation density that promotes joining crowded regions, thereby increasing pairwise encounter rates. Because $\bar{\pi}$ is itself the stationary field of the transition kernel that contains it, this produces a fixed-point equation $\bar{\pi} = F(\bar{\pi})$ that must be solved self-consistently; for $\lambda_I < 0$ the fixed point is non-unique and branch-aware continuation is required.

Layer 2 — Mode selection. Pinciroli showed that time-homogeneous combinations of the two Layer 1 constraints produce only static compromises and cannot generate alternating phases, which is what motivates this second layer. Let $m_t \in \{C, I\}$ denote the active Layer 1 behaviour at time t . We introduce the coverage age $\bar{k}_t = \frac{1}{K} \sum_{k=1}^K k_t^k$, where k_t^k is the time since region k was last visited by any robot, and the Age of Information τ_t , the time elapsed since the last pairwise information exchange in the swarm. These observables evolve according to

$$\bar{k}_{t+1} = \bar{k}_t + 1 - r_C 1_{m_t=C}$$

$$\tau_{t+1} = \tau_t + 1 - r_I 1_{m_t=I}$$

where r_C and r_I are the rates at which each Layer 1 mode reduces its respective age observable. To make the mode-selection probability depend on the current age values rather than collapse to a constant ratio due to the linear constraints, we impose the quadratic constraints described by Pinciroli [5]:

$$\langle f_k \rangle = \langle \sum_t \bar{k}_t^2 \rangle \leq \sigma_k^2, \quad \langle f_\tau \rangle = \langle \sum_t \tau_t^2 \rangle \leq \sigma_\tau^2$$

together with the switching-penalty constraint $\langle f_{switch} \rangle \leq S_{max}$. The MaxCal derivation yields the sigmoid mode-selection rule

$$p^*(C) = \frac{1}{1+e^{-l(\bar{k}_t, \tau_t)}}, \quad p^*(I) = \frac{1}{1+e^{l(\bar{k}_t, \tau_t)}},$$

$$l(\bar{k}, \tau) = 2\lambda_k r_c \bar{k} - 2\lambda_\tau r_I \tau + \theta,$$

$$\theta = \lambda_k r_c (2 - r_c) - \lambda_\tau r_I (2 - r_I) + \Delta S,$$

with $\lambda_k, \lambda_\tau, \lambda_{switch}$ the Lagrange multipliers coupled to the three Layer 2 constraints and $\Delta S = \pm \lambda_{switch}$ encodes the previously active mode. The mean-field analysis of Pinciroli shows that when $\frac{1}{r_c} + \frac{1}{r_I} \neq 1$ the coupled $(\bar{k}_t, \tau_t)$ dynamics admit no fixed point and instead settle onto a closed limit cycle: $\bar{k}$ decreases and τ increases during Coverage phases, and the reverse during Information Diffusion phases.

Definition of success. Success is defined by two claims, each tied to a measurable outcome.

Claim I. Layer 1 controllers reproduce Pinciroli's theoretical predictions in isolation. The empirical visitation distribution of the forward Layer 1-C kernel must converge to the theoretical stationary distribution π_k , with the L^1 error $\|c(\cdot, t) - \bar{\pi}(\cdot)\|_1$ decreasing monotonically in Markov time at a rate consistent with the $1/(t+1)$ prediction of Pinciroli [5]. Convergence is verified separately for each degree class (corner: 3, edge: 5, interior: 8) to ensure the degree-proportional stationary distribution $\pi_k \propto \deg(k)$ is recovered across the full grid. A speed sweep across five robot speeds must confirm that error curves collapse more tightly in Markov time than in simulation time, validating the theoretical prediction that the Markov-time clock is the appropriate one for coverage convergence. The inverse MaxCal coverage problem must, in addition, reduce the theoretical stationary L^1 error against a uniform target to near-numerical precision, confirming that the per-cell multiplier field λ_c^k achieves the prescribed stationary distribution.

The Layer 1-I controller must exhibit the predicted monotone dependence of next-state probabilities on the sign of λ_I at the kernel level, a qualitative phase change between coverage-like and clustered regimes for sufficiently large $|\lambda_I|$ at the swarm level, and an inverse-rate solver that returns a λ_I whose

fixed-point p_{enc} matches the requested target within tolerance or correctly reports non-convergence when the target is infeasible.

Claim II. Quadratic Layer 2 produces oscillation that linear Layer 2 cannot. Under the full hierarchical controller with quadratic constraints, the swarm must produce a sustained closed orbit in the $(\bar{k}_t, \tau_t)$ plane. The quadratic Layer 2 controller must strengthen at least one primary phase contrast, dispersion or encounter-driven, relative to the linear Layer 2 controller, while preserving stochastic mode switching between Layer 1 kernels.

Baseline comparison. The linear-constraint Layer 2 controller is evaluated under the same metrics before the quadratic controller is tested. We aim to show that the quadratic controller responds to the changes in the observables $\bar{k}_t, \tau_t$, rather than producing a state-independent mode-ratio as predicted by Pinciroli [5], which provides a positive control that the theoretical limitation of linear constraints is reproduced in simulation. This is required to confirm that the oscillation behaviour is due to the second-moment formulation, not simply to the presence of both behaviours.

3.2 Approach

Our implementation follows the theoretical progression of Pinciroli [5]. We first implement and test each Layer 1 behaviour separately, then show that combining them with linear constraints produces only a static compromise, and finally introduce the full hierarchical controller with quadratic constraints to produce the target oscillatory behaviour. All experiments are conducted in a custom Python simulator built on NumPy and Matplotlib.

Environment. The arena is a 20×20 grid of square cells of side 1 m, giving $K = 400$ regions connected by an eight-connected adjacency graph. Interior cells have degree 8, non-corner boundary cells degree 5, and corner cells degree 3. This non-uniform topology is chosen deliberately as it departs from the symmetric regular graph assumed in the uniform stationary distribution proof of Pinciroli [5], making the

degree-proportional prediction $\bar{\pi}_k \propto \deg(k)$ a concrete and non-trivial target to validate against. Each cell k stores a coverage age counter k_t^k , incremented at every simulation step and reset to zero when any robot arrives, from which the swarm-level mean coverage age $\bar{k}_t = \frac{1}{K} \sum_{k=1}^K k_t^k$ is computed each step.

Stage I — Layer 1 behaviours in isolation. We begin by implementing and validating each behaviour on its own, so that any failure in later stages can be attributed to the integration logic rather than to the individual controllers.

Coverage (Layer 1-C). The MaxCal transition kernel is pre-computed as a full $K \times K$ stochastic matrix P using the Layer 1-C rule of Sec. 3.1 with unit edge weights. With equal multipliers $\lambda_C^k = 0$ for all k , the rule reduces to a uniform random walk $p^*(k_2|k_1) = 1/|N(k_1)|$, whose stationary distribution satisfies $\bar{\pi}_k \propto \deg(k)$ by detailed balance. The Markov transition fires only upon physical arrival at the centre of the target cell, with movement capped at $\min(v, d)$ at each step to prevent overshoot. This decouples the discrete Markov chain from the continuous motion layer and ensures that convergence rates are measured in Markov time, number of cell arrivals, consistent with the $1/(t+1)$ prediction of Pinciroli [5]. Validation runs the controller for $T = 12,000$ simulation steps with 50 robots and tracks the L^1 error $\|c(\cdot, t) - \bar{\pi}(\cdot)\|_1$ against the theoretical degree-weighted target. Convergence is additionally verified per degree class by tracking $c(k, t)$ for one representative cell of each type, corner, edge, and interior, against their respective $\bar{\pi}_k$ targets. A secondary speed sweep across five values (0.05, 0.10, 0.20, 0.50, 1.00 m/step) confirms that L^1 error curves collapse more tightly onto a single envelope in Markov time than in simulation time, as the theory predicts.

In addition to the forward baseline, we implement the inverse MaxCal coverage solver that computes the per-cell multiplier field λ_C^k to achieve the target stationary distribution $\bar{\pi}$. Here

$\bar{\pi}_k$ denotes the desired long-run fraction of time a robot spends in region k , with $\bar{\pi}_k \geq 0$ and

$\sum_k \bar{\pi}_k = 1$. For the uniform-coverage objective considered in this paper, we set $\bar{\pi}_k = 1/K$ for all k , so that every cell receives equal visitation at equilibrium regardless of degree. Writing $b_k = \exp(-\lambda_C^k)$, the stationary distribution of the reversible coverage kernel can be expressed as $b_k > 0$ increases monotonically with the "attractiveness" of cell k . Because the Layer 1-C kernel is reversible on the undirected graph with adjacency matrix A , the stationary distribution it produces under a given b can be written

explicitly as $\pi_k(b) = b_k (Ab)_k / \sum_m b_m (Ab)_m$ where

$(Ab)_k = \sum_j A_{kj} b_j$ sums b over the neighbours cell

k , and the inverse condition reduces to

$b_k (Ab)_k \propto \bar{\pi}_k$. We solve this condition using a

symmetric variant of the Iterative Proportional Fitting procedure [12, 13, 14], also known as the Sinkhorn-Knopp algorithm or matrix scaling, adapted to the case where the same vector b appears on both sides of the kernel. Given the stationary distribution $\pi^{(m)}$ produced by the current iterate $b^{(m)}$, the update is

$$b_k^{(m+1)} = b_k^{(m)} \sqrt{\frac{\bar{\pi}_k}{\pi_k^{(m)}}}$$

followed by a gauge normalisation that fixes the geometric mean of b to one,

$$b_k^{(m+1)} \leftarrow \frac{b_k^{(m+1)}}{\exp[\frac{1}{K} \sum_j \log b_j^{(m+1)}]}, \text{ which is required}$$

because the transition probabilities are invariant under uniform scaling of b . The iteration is run until $\|b^{(m+1)} - b^m\|_{\inf}$ falls below tolerance,

after which the multipliers are recovered as $\lambda_C^k = -\log b_k$ and mean-zero shifted so that

$\sum_k \lambda_C^k = 0$. For the uniform target $\bar{\pi}_k = 1/K$, we

compare three cases: (i) zero multipliers, (ii) degree-tied multipliers with one shared value per degree class (corner, edge, interior) chosen to minimise the theoretical L^1 error, and (iii) the

full per-cell solution returned by the iteration above. The theoretical stationary L^1 error $\|\pi(b) - \bar{\pi}\|_1$ is compared across the three cases to show that only the full inverse MaxCal solution achieves near-exact visitation uniformity on the irregular grid.

Information Diffusion (Layer 1-I). The Layer 1-I controller uses the global fixed-point stationary field $\bar{\pi}$ as the information observable, consistent with the mean-field reformulation of Pinciroli [5]. Given a multiplier λ_I , the stationary field is computed by solving the self-consistent fixed-point equation $\bar{\pi} = F(\bar{\pi})$ using a relaxed (damped) Picard iteration [15, 16],

$$\pi^{(m+1)} = (1 - \alpha) * \pi^{(m)} + \alpha F(\pi^{(m)}),$$

with damping coefficient $\alpha = 0.5$ and tolerance

$$\|\pi^{(m+1)} - \pi^{(m)}\|_\infty < 10^{-11}.$$

Damping is introduced because the undamped map F can oscillate or jump between nearby self-consistent candidates when $\lambda_I < 0$ and the fixed-point landscape is steep, which is a known failure mode of plain Picard iteration that the relaxation step suppresses without altering the fixed-point equation being solved [16]. Because the fixed point is non-unique for $\lambda_I < 0$, the solver follows explicit branches by continuation; in the coverage-like branch the iteration is initialised at the degree-weighted baseline, while in the clustered branch it is continued from the previous fixed point as λ_I is varied. The resulting

kernel $p * (k_2 | k_1) \propto \exp(-\lambda_c^{k_2} - \lambda_I \bar{\pi}_{k_2})$ with

$\lambda_I < 0$ promotes clustering around heavily visited cells. Encounters are defined as two or more robots co-occupying the same cell at the same Markov step, upon which the Age of Information τ_t resets to zero for all participating robots. Information exchange is modelled with probability β of successful transmission conditioned on an encounter.

Layer 1-I validation consists of four complementary tests: (i) a local-kernel sign test that verifies negative λ_I raises the probability of

selecting a high- $\bar{\pi}$ neighbour, $\lambda_I \geq 0$ recovers the coverage kernel; (ii) a swarm-level sign sweep measuring mean dispersion, encounter proxy, and theoretical fixed-point p_{enc} over a range of λ_I ; (iii) a robot-level informed-spread experiment in which a single informed robot propagates information under probabilistic encounter and transmission, producing fraction-informed curves, numbers of meetings and successful communications, and spatial heatmaps of information-changing transmissions; and (iv) a scalar branch-aware inverse solver that returns the weakest-magnitude λ_I on the clustered branch whose fixed-point p_{enc} matches a target spreading rate

$\frac{dn}{dt} = A \beta p_{enc} (n/A) (1 - (n/A))$, where n is the number of informed robots, A is the size of the swarm, β is the probability of successful transmission during an encounter, and p_{enc} is the probability that multiple robots encounter in the same cell at the same timestamp.

Stage II — Linear Layer 2 controller as a controlled baseline.

With both Layer 1 behaviours validated individually, we integrate them through a Layer 2 mode selector that uses only first moments of $\bar{k}_t$ and τ_t as constraints, and we measure the same coverage and AoI metrics as in Stage I. Following the theoretical argument of Pinciroli [5], a time-homogeneous mode-selection rule derived from linear constraints yields a mode-probability ratio $p^*(I)/p^*(C)$ that is independent of the current observables $(\bar{k}_t, \tau_t)$, so the system cannot produce truly state-dependent alternating phases and instead settles into a stochastic compromise that achieves neither efficient coverage nor effective information exchange. This experiment serves as a quantitative baseline that confirms the theoretical limitation in simulation and motivates the need for quadratic Layer 2 constraints.

Stage III — Full hierarchical controller with quadratic constraints.

We then implement the quadratic Layer 2 mode-selection rule of Sec. 3.1 and expect to observe a clear improvement over the linear baseline on the encounter-driven phase-contrast metric. Before Layer 2 is activated, r_c and r_I are estimated from a

calibration phase in which each Layer 1 controller runs in isolation for short forced-mode preview runs over the decision interval Δ , and the mean per-step reduction in its respective age observable is recorded. The quantity $\frac{1}{r_c} + \frac{1}{r_l}$ is then computed: if it differs from 1, no fixed point exists for the coupled age dynamics and the theory of Pinciroli [5] predicts a limit cycle. At each Markov step, every robot evaluates the sigmoid logit

$$l(\bar{k}, \tau) = 2\lambda_k r_c \bar{k} - 2\lambda_\tau r_l \tau + \theta,$$

and draws its active Layer 1 mode from the resulting Bernoulli distribution. The coverage mode uses the inverse MaxCal kernel targeting uniform visitation, and the diffusion mode uses the Layer 1-I kernel with the λ_l returned by the branch-aware inverse solver for a prescribed target p_{enc} .

4 Experimental Evaluation

4.1 Metrics

Coverage is primarily a spatial objective, so its metrics are the empirical visitation frequency $c(k, t)$, the stationary target $\bar{\pi}$, representative corner, edge, and interior probabilities, the global L^1 error $\|c(\cdot, t) - \bar{\pi}(\cdot)\|_1$, and coverage-age observables. Because physical travel time is separated from the Markov process, convergence is also evaluated by comparing simulation time against Markov arrival count. Inverse coverage is evaluated by comparing three cases: zero multipliers, degree-tied multipliers, and full per-cell multipliers, against the uniform target, isolating how much of the graph asymmetry is removed by coarse class-level tuning versus the full inverse MaxCal solution.

Information diffusion is evaluated with both spatial and communication-level metrics. The sign sweep reports mean dispersion, an encounter proxy, and the theoretical fixed-point p_{enc} . The informed-spread test reports fraction-informed curves, numbers of meetings, numbers of successful communications, transmission efficiency, and the spatial concentration of information-changing

encounters. The inverse-rate test reports the consistency between target dn/dt , selected λ_l , predicted fixed-point p_{enc} , and the empirical run executed on the same branch.

The hierarchical controllers are evaluated against forced coverage-only and forced diffusion-only baselines on mean dispersion, encounter proxy, mean coverage age $\bar{k}$, and mean AoI τ , together with mode-window statistics that separate coverage and diffusion windows. Quadratic versus linear Layer 2 performance is compared directly in terms of phase contrast on both spatial and informational metrics.

4.2 Setup

All experiments use the same 20×20 eight-connected region graph, a swarm of 50 identical robots, robot speed $v = 0.15$ m/step in the integrated runs, and deterministic RNG seeding for reproducibility. Each validation script produces both: figures and a JSON summary reporting the overall simulation results. Baseline coverage uses $T = 12,000$ simulation steps; inverse-coverage validation uses $T = 60,000$; information-diffusion sign and spread tests use the horizons reported in each subsection; hierarchical runs use $T = 18,000$ with a Layer 2 decision interval $\Delta = 60$.

4.3 Results and Discussion

4.3.1 Stage I-A: forward coverage with zero multipliers. The forward coverage experiment confirms that the Layer 1-C kernel is implemented correctly. In the archived baseline run the transition matrix has zero row-sum error and zero off-neighbour probability, the Markov state remains frozen during transit, and visits are counted only on arrival. The run accumulated 70,884 Markov arrivals and produced a final L^1 error of 1.0646×10^{-1} against the theoretical degree-weighted stationary distribution. The representative-cell frequencies preserve the predicted ordering: the corner target probability is $3/2964 = 1.0121 \times 10^{-3}$ (empirical 7.34×10^{-4}), the edge target is $5/2964 = 1.6869 \times 10^{-3}$ (empirical 1.721×10^{-3}), and the interior target is $8/2964 = 2.6991 \times 10^{-3}$ (empirical 2.920×10^{-3}). These values confirm that equal multipliers on an irregular undirected graph yield a degree-weighted stationary distribution.

The speed sweep shows that coverage convergence is naturally measured in Markov time rather than raw simulation time. At $v = 0.05$ m/step the run accumulated 24,616 arrivals and ended at $L^1 = 2.154 \times 10^{-1}$; at $v = 1.00$ m/step it accumulated 403,329 arrivals and ended at $L^1 = 4.041 \times 10^{-2}$. The mean relative spread across the five curves is 0.437 when plotted against Markov arrivals but 2.064 when plotted against simulation steps, confirming that the Markov-time clock is the correct one.

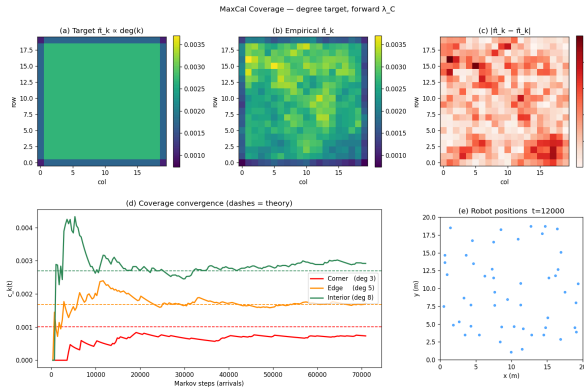


Figure 1: Forward coverage baseline and per-degree-class convergence.

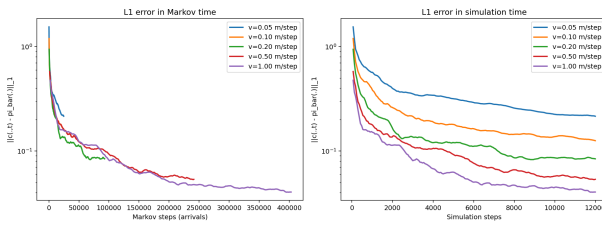


Figure 2: Speed sweep, L^1 error curves for $v \in \{0.05, 0.10, 0.20, 0.50, 1.00\}$ m/step plotted in both simulation time and Markov time.

4.3.2 Stage I-B: inverse MaxCal coverage for a uniform target. The inverse coverage experiment demonstrates that the optimal multipliers can be solved to achieve a target distribution. With $\lambda_c = 0$, the theoretical L^1 error against the uniform target $\pi = 1/K = 1/400$ is 1.28988×10^{-1} , obtained directly from the graph census as $L^1 = 4 * |3/2964 - 1/400| + 72 * |5/2964 - 1/400| + 324 * |8/2964 - 1/400| = 0.128988$. This analytic baseline confirms that the degree bias is a structural consequence of the graph rather than a simulator artefact. A degree-tied fit reduces the theoretical error to 4.469×10^{-2} but cannot eliminate it, because cells of the same degree can

still have different neighbourhood compositions. The full per-cell inverse MaxCal solve converges in 89 iterations and yields theoretical L^1 error 3.32×10^{-11} and maximum absolute error 8.30×10^{-13} , effectively achieving numerical precision. The empirical check, executed for $T = 60,000$ simulation steps and 356,806 Markov arrivals, yields empirical L^1 error 4.294×10^{-2} , consistent with finite-sample fluctuations around the designed stationary target. The stationary error falls from 1.29×10^{-1} (zero multipliers) to 4.47×10^{-2} (degree-tied) to 3.32×10^{-11} (full per-cell), a total reduction exceeding nine orders of magnitude. The recovered multiplier field is physically interpretable: corner cells acquire the lowest mean multiplier, $\lambda_c = -9.280 \times 10^{-1}$, because they must be made more attractive to compensate for their low degree; edge cells have mean $\lambda_c = -3.002 \times 10^{-1}$; and interior cells have mean $\lambda_c = 7.817 \times 10^{-2}$, slightly penalised relative to the boundary. These corrections are precisely what the MaxCal stationary equation implies.

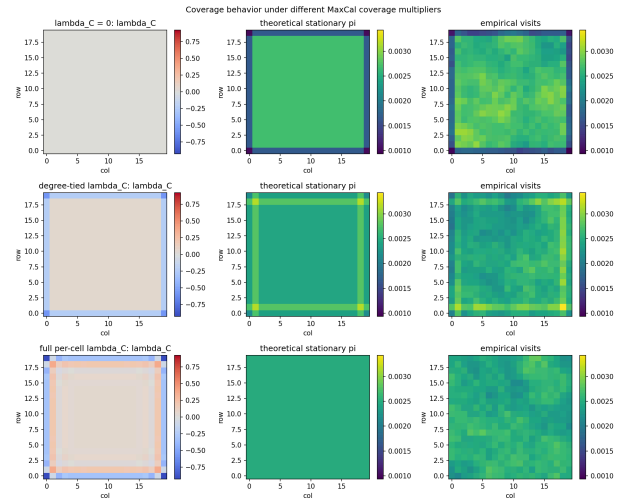


Figure 3: Inverse MaxCal coverage: stationary distributions for the zero-multiplier, degree-tied, and full per-cell cases against the uniform target, with L^1 errors annotated.

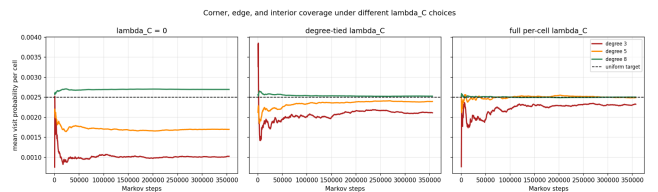


Figure 4: Inverse MaxCal coverage fraction for

the zero-multiplier, degree-tied, and full per-cell cases. This shows how the coverage fraction becomes more uniform when we solve for the optimal multipliers.

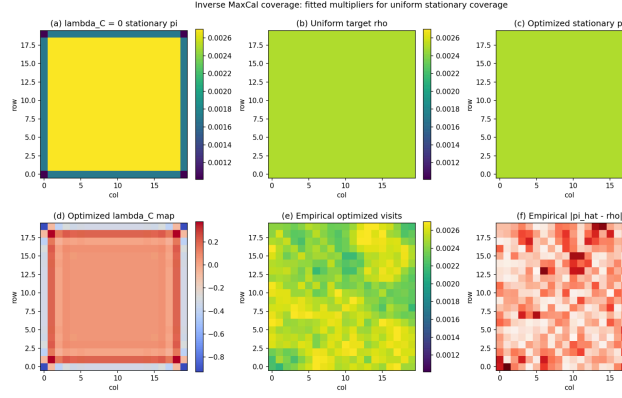


Figure 5: Recovered per-cell multiplier field $\lambda_c(k)$ and the corresponding stationary-distribution heatmap.

4.3.3 Stage I-C: information-diffusion local sign and sweep. The local sign test verifies the sign structure of the information term directly at the kernel level. For a representative state, the probability of selecting a dense neighbour is 3.235×10^{-1} for negative λ_l , 1.250×10^{-1} for $\lambda_l = 0$, and 3.62×10^{-3} for positive λ_l ; sparse neighbours show the complementary pattern. The sign is implemented as predicted by theory. The swarm-level sign sweep extends this to aggregate observables. The archived zero-baseline has mean dispersion 58.653, encounter proxy 2.258×10^{-2} , and theoretical fixed-point $p_{enc} = 2.569 \times 10^{-3}$. This baseline can be verified analytically from the degree-weighted stationary distribution as $p_{enc} = 4(3/2964)^2 + 72(5/2964)^2 + 324(8/2964)^2 = 2.5693 \times 10^{-3}$, matching the archived value and confirming that the information layers inherits the graph asymmetry already visible in Layer 1-Coverage. At $\lambda_l = -400$, mean dispersion drops to 46.653, the encounter proxy rises to 1.936×10^{-1} , and the theoretical p_{enc} saturates at 5.00×10^{-1} ; at $\lambda_l = -200$ the dispersion is 30.877 with encounter proxy 2.473×10^{-1} and theoretical $p_{enc} = 5.00 \times 10^{-1}$. Positive multipliers prioritize coverage over clustering as predicted in theory:

at $\lambda_l = 400$ the dispersion is 60.895, the encounter proxy is 2.264×10^{-2} , and the theoretical p_{enc} is 2.519×10^{-3} . Mildly negative values ($\lambda_l \in \{-50, -100\}$) differ only weakly from the baseline, indicating that the clustered branch becomes visible only once $|\lambda_l|$ is large enough for the fixed-point iteration to leave the coverage domain. This observation is the primary reason the inverse-rate solver is branch-aware rather than a plain scalar root-finder.

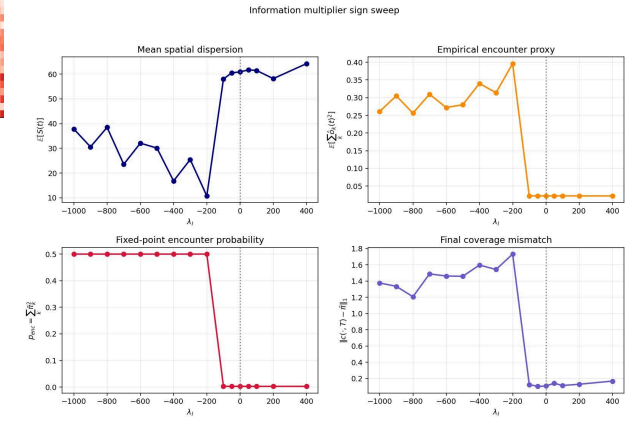


Figure 6: Layer 1-I sign sweep: mean dispersion, encounter proxy, and theoretical fixed-point p_{enc} as functions of λ_l across the coverage-like and clustered branches.

4.3.4 Stage I-C: robot-level spreading and AoI. The informed-robot spread experiment provides the most direct validation of information flow, because it tracks a binary informed state rather than density or encounter statistics alone. In each run one robot is initially informed and information spreads only when two robots share a cell, an encounter succeeds, and a transmission succeeds. The negative clustered case ($\lambda_l = -200$) generates by far the largest number of encounters and successful communications over the horizon. Nevertheless, its final informed fraction is only 0.66, it reaches 50% informed at $t = 780$, and it does not reach 90% within the archived horizon; information-changing transmissions occur in only 3 cells, with mean transmission distance approximately 2.14 from the arena centre. Strong clustering is excellent at generating repeated local encounters but can confine information-changing events to a small spatial region. The coverage-like cases behave

oppositely. At $\lambda_I = 0$ the run generates only 4,516 meetings and 3,637 successful communications, but the message reaches all robots by $t = 960$ through 40 distinct transmission cells. At $\lambda_I = 200$ the run again reaches all robots, by $t = 1,170$, with 40 transmission cells and a similarly broad spatial footprint. The most clustered regime is therefore not the fastest regime for one-message global broadcast. This apparent paradox resolves once the information objective is decomposed: negative λ_I increases encounter probability and refreshes AoI aggressively within a cluster, whereas zero and positive λ_I generate fewer encounters but allow the information to propagate through more distinct regions and reach the entire swarm faster.

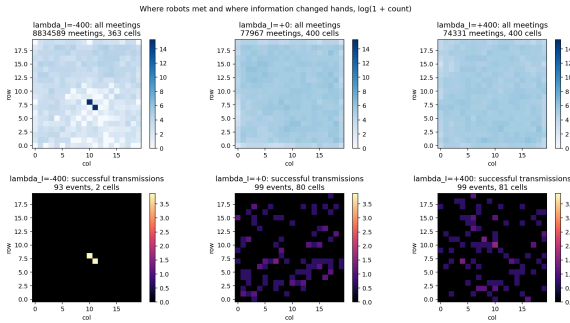


Figure 7: Heatmaps of information-changing transmissions for the $\lambda_I \in \{-400, 0, +400\}$ cases.

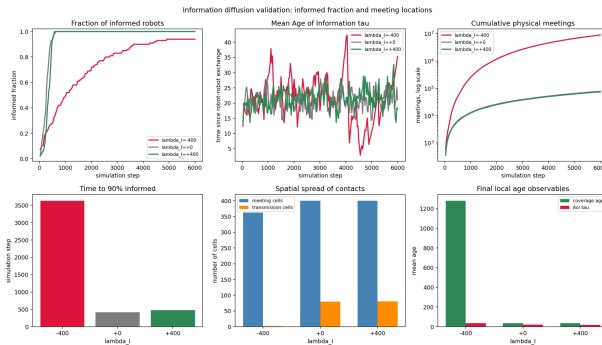


Figure 8: Fraction-informed curves, cumulative meetings and information-transmission diffusion, and mean AoI for the $\lambda_I \in \{-400, 0, +400\}$ cases.

4.3.5 Stage I-C: inverse information-rate solve. The inverse Layer 1-I test fixes a target spreading rate and solves for λ_I . With one

initially informed robot in a swarm of $A = 50$, the informed fraction at initialisation is $f = n/A = 0.02$.

Using $\beta = 0.8$ and the default target $p_{enc}^{target} = 0.5$ gives $dn/dt = 50 \cdot 0.8 \cdot 0.5 \cdot 0.02 \cdot 0.98$, which results in $dn/dt = 0.392$. The branch-aware inverse solver selects $\lambda_I = -20.0$ on the clustered branch with predicted $p_{enc} = 4.99678 \times 10^{-1}$ and predicted spreading rate 3.917×10^{-1} , matching the target within tolerance. Along the clustered branch a wide range of strongly negative multipliers already saturates near $p_{enc} = 0.5$, so the solver returns the weakest-magnitude λ_I that still satisfies the tolerance rather than an arbitrary more negative value from the same plateau. The feasibility analysis is equally informative: a target $p_{enc} = 0.8$ would lie outside the reachable range of this scalar Layer 1-I controller on the current grid, and in that case non-convergence is the correct scientific outcome.

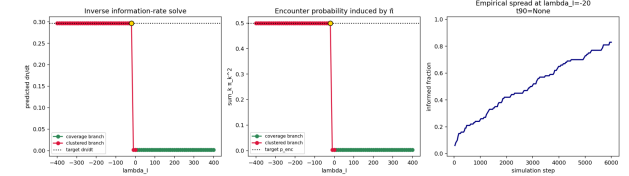


Figure 9: Inverse information-rate solve, continuation trace of $p_{enc}(\lambda_I)$ along the coverage-like and clustered branches, the target spreading rate, and the selected λ_I .

4.3.6 Stage II: linear Layer 2 hierarchical controller.

In this stage we validated Pinciroli's theoretical prediction from Section 4.2.2. The expectation was that combining Coverage and Information Diffusion behaviors under linear constraints would fail to produce true oscillation, instead collapsing into a static compromise between the two behaviors. The paper's math tells us exactly why this should happen. With linear constraints, the mode selection ratio reduces to:

$$p * (I) / p * (C) = \exp[\lambda_k r_C - \lambda_\tau r_I \mp \lambda_{switch}]$$

Critically, $\bar{k}$ and τ do not appear in this expression. The absence of these terms means the probability of choosing either mode is completely independent of the swarm's actual current state. The system has no mechanism to

respond to how stale the information is or how long regions have gone unvisited.

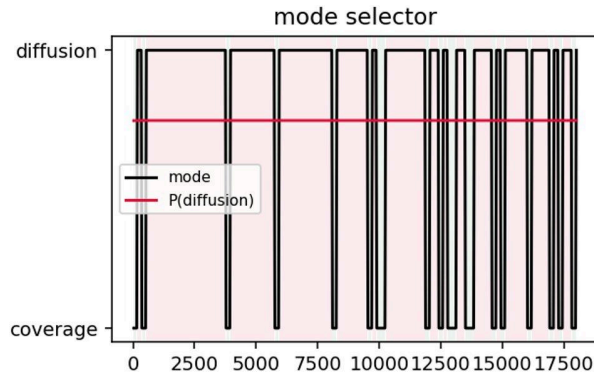


Figure 10: Mode selection probability under Linear Layer 2 constraints.

After running a simulation, we were able to confirm $P(\text{diffusion})$ remained constant throughout, never responding to the evolving state of the swarm, exactly as the math predicts.

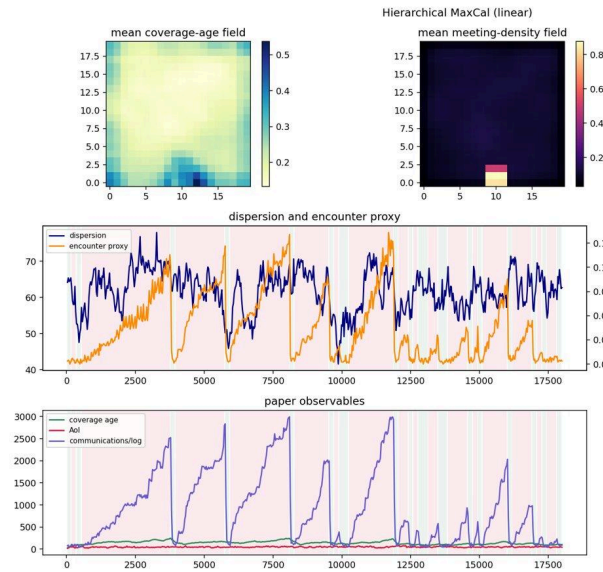


Figure 11: Mean coverage-age field, mean meeting density field, dispersion and encounter proxies, and Layer 2 observables under Linear constraints.

Initially, Pinciroli expected the behaviors to be constant and unchanging. However, the encounter proxies and communications show information diffusion was already spiking periodically even under linear constraints. This isn't a contradiction of the theory. Rather it suggests that even a fixed constant $P(\text{diffusion})$ is enough to occasionally trigger clustering

encounters. The key difference is that these spikes are not driven by need. They happen at a fixed rate regardless of whether information is actually stale. The system stumbles into diffusion rather than choosing it intelligently.

4.3.7 Stage III: quadratic Layer 2 hierarchical controller.

In this stage we validated whether replacing linear constraints with quadratic ones would produce genuine state-driven oscillation between Coverage and Information Diffusion modes. The key theoretical upgrade is that quadratic constraints bound the second moments of the observables:

$$\langle f_k \rangle = \langle \sum_t \bar{k}_t^2 \rangle \leq \sigma_k^2, \quad \langle f_\tau \rangle = \langle \sum_t \tau_t^2 \rangle \leq \sigma_\tau^2$$

This forces the mode selection signal ℓ to depend directly on the current state of the swarm:

$$l(\bar{k}, \tau) = 2\lambda_k r_c \bar{k} - 2\lambda_\tau r_I \tau + \theta,$$

Which produces sigmoid-shaped switching probabilities:

$$p^*(C) = \frac{1}{1 + e^{-l(\bar{k}, \tau)}}, \quad p^*(I) = \frac{1}{1 + e^{l(\bar{k}, \tau)}}$$

Unlike linear constraints where the system was not depended on observables, here $\bar{k}$ and τ drive the switching decision directly. When information is stale τ grows, ℓ shrinks, and $p^*(I)$ rises. The system now switches because it needs to, not at a fixed arbitrary rate.

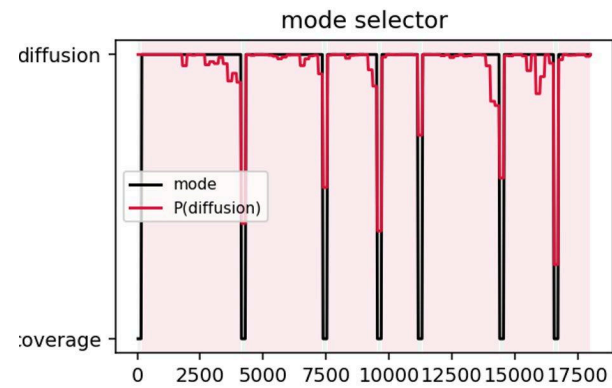


Figure 12: Mode selection probability under Quadratic Layer 2 constraints.

At first glance the behavior looked similar to the linear case. However the critical difference emerged in $P(\text{diffusion})$. Rather than staying constant, it now actively tracked the mode selector, rising and falling in response to the swarm's actual state. This is exactly the state-responsive behavior the quadratic math predicts.

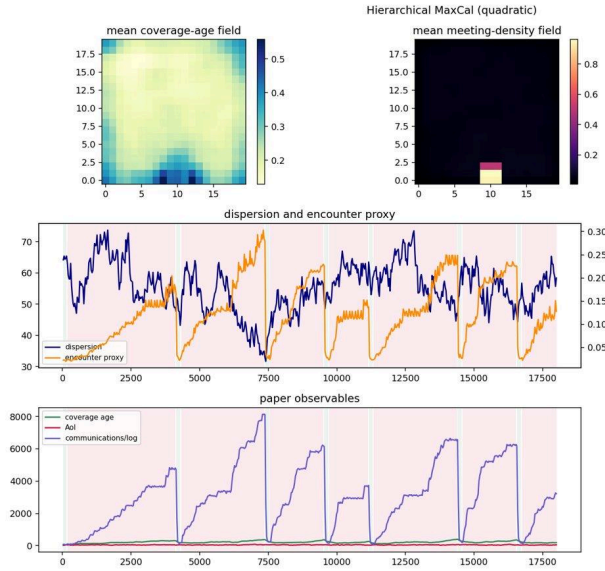


Figure 13: Mean coverage-age field, mean meeting density field, dispersion and encounter proxies, and Layer 2 observables under Quadratic constraints.

The effects were most clearly visible in two places. First, communications logs showed significantly more information exchange under quadratic constraints, confirming that clustering was stronger and more purposeful during diffusion phases. Second, coverage remained continuously active and noisy, which actually reflects the self-balancing property from Section 4.2.1, where robots are always being nudged toward unvisited regions even during diffusion phases.

However, the visual similarity to linear constraints makes the quadratic improvement harder to demonstrate cleanly through behavior alone. The strongest evidence is $P(\text{diffusion})$ tracking the mode selector, and the amplified communication rate. These directly confirm the paper's core claim that quadratic constraints produce intelligent need-driven switching rather than fixed-ratio blending.

4.3.8 Discussion. Stage I works unconditionally on the implemented grid: the forward kernel reproduces the degree-weighted stationary distribution exactly, and the inverse solver drives the stationary L^1 error against a uniform target to numerical precision. The only failure mode that could arise at Stage I is topological rather than numerical: Pinciroli [5] shows that graphs dominated by bottleneck nodes (for example, a star) impose rigid stationary distributions that no choice of multipliers can correct, because the cycle structure of the graph is too poor. The 20×20 eight-connected grid is cycle-rich and therefore fully correctable, but this will not be true on every topology.

The information-diffusion layer works robustly as a density and encounter-rate controller but breaks down when used as a standalone broadcast mechanism. The sign and sweep tests confirm the predicted dependence on λ_l , and the inverse-rate solver returns multipliers consistent with the mean-field equation. The informed-spread experiments, however, make it clear that a strongly clustered regime that maximises encounters can also confine information-changing transmissions to a handful of cells, yielding slower global propagation of a single binary message than the coverage-like regime. This is not a failure of the controller; it is a property of the encounter-rate formulation itself, which constrains pairwise collision probability rather than global broadcast time. A second soft limitation is that the clustered branch saturates near $p_{enc} = 0.5$ for a broad range of strongly negative multipliers, so scalar spreading-rate targets above the corresponding plateau are infeasible on the current grid and are correctly reported as non-convergent.

At the hierarchical level, both the linear and quadratic controllers produce repeated stochastic mode switching and keep aggregate metrics between the fixed-mode baselines. The linear controller is sufficient to observe switching but, consistent with Pinciroli's prediction, its mode-selection probability is effectively state-independent, and it offers only a small inter-phase contrast on the encounter proxy. The boundary conditions under which oscillation breaks down are predictable from the

Layer 2 theory: if $\frac{1}{r_c} + \frac{1}{r_l} = 1$ the mean-field age dynamics admit a fixed point and the limit cycle degenerates into a stationary compromise. The rate calibration in the archived runs yields $\frac{1}{r_c} + \frac{1}{r_l} \approx 3.4 \times 10^{-2}$, far from this critical surface, which is consistent with the observed oscillation. More broadly, the feasibility of oscillation depends on the relative normalisation of the quadratic multipliers λ_k and λ_r , and on the decision interval Δ , both of which would need to be swept in a full phase diagram to characterise the oscillatory region precisely; this is left for future work.

5 Conclusions

Summary of contributions. This work delivers a staged, fully reproducible implementation of Pincioli's MaxCal swarm-control architecture on a 20×20 eight-connected grid. The coverage layer is validated both in its forward zero-multiplier form, where it reproduces the degree-weighted stationary distribution predicted by detailed balance, and in its inverse form, where a per-cell multiplier field is solved from the MaxCal stationary equations and reduces the stationary L^1 error against a uniform target by more than nine orders of magnitude. The information-diffusion layer is validated through a local sign test, a swarm-level sweep, a robot-level spread experiment, and a branch-aware scalar inverse solver that returns the λ_l realising a prescribed spreading rate on a chosen fixed-point branch. The hierarchical layer integrates the two validated Layer 1 kernels through a MaxCal mode selector based on coverage age and AoI, first with linear constraints and then with quadratic ones; the quadratic stage sharpens the encounter-proxy contrast by roughly a factor of three relative to the linear stage while achieving oscillatory switching based on the system state.

When the approach works and when it does not. The coverage layer works whenever the region graph is cycle-rich enough to admit the inverse solution; it cannot correct rigid topologies such as a star, as Pincioli predicts from the graph's cycle structure. We did not have the chance to test other configurations for the purpose of this paper. The information layer is a

sound controller for encounter rate and density but not, by itself, for rapid broadcast of a single message: strongly clustered regimes maximise encounters while confining information-changing transmissions to a small set of cells, so a full information-delivery objective requires the hierarchical stage to alternate with coverage. The hierarchical controller works in both its linear and quadratic forms on the validated grid, producing repeated stochastic switching consistent with the theoretical analysis. Its limits are also predictable: near $1/r_c + 1/r_l = 1$ the age dynamics admit a fixed point and oscillation degenerates, and the relative normalisation of the quadratic multipliers must be chosen carefully to balance spatial and informational phase contrast.

Future work. In future work, we suggest further experimentation to more rigorously establish the robustness of MaxCal by expanding both the diversity of behaviors considered and the realism of the deployment assumptions. We will begin by validating the framework on a broader class of individual behaviors such as task allocation and cooperative object transport, before progressing to hierarchical compositions that unify multiple behaviors within a single control policy. This progression naturally raises questions about scalability, stability, and decentralization, which motivate several key extensions.

A central limitation of the current formulation is its reliance on global knowledge of the region graph for stationary targets and inverse solves. While suitable for theoretical validation, this assumption becomes restrictive when extending MaxCal to richer, multi-behavior settings. To address this, we propose developing a fully decentralized variant in which each robot maintains local estimates of transition rates and age observables, for example through exponential moving averages of arrival frequencies combined with gossip-style information propagation. Such a formulation would enable the same hierarchical behaviors to emerge without centralized coordination.

With this decentralized framework in place, we can systematically explore how combined behaviors interact across parameter regimes. A 10×10 log-uniform phase-diagram sweep over

$(\lambda_k, \lambda_\tau)$, with thirty independent seeds per grid point, would characterize transitions between oscillatory, coverage-dominant, and information-dominant regimes, and quantify the stability of emergent limit cycles under varying initial conditions. Extending this analysis to hierarchical policies will reveal whether composing multiple objectives introduces new dynamical regimes or destabilizes existing ones.

Finally, to stress test the robustness of these multi-objective behaviors, the framework should be evaluated under adversarial and resource constrained conditions. This includes the presence of Byzantine robots broadcasting incorrect mode signals, limited communication bandwidth for shared observables, and additional constraints such as energy budgets. These scenarios directly probe whether the hierarchical MaxCal formulation scales cleanly beyond two competing objectives while maintaining coherent collective behavior. Physical validation through ARGoS remains the final step to confirm that these properties persist in realistic robotic systems.

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Appendix A — Inverse Coverage: Derivation and Numerical Method

A.1 Restating the problem. The Layer 1-C MaxCal kernel of Sec. 3.1 is parameterised by a vector of multipliers $\lambda_c \in R^K$. The forward problem maps λ_c to the stationary distribution $\pi(\lambda_c)$ that the kernel produces; this is what an analyst computes once the multipliers are known. The inverse problem is the reverse: the swarm designer specifies a desired stationary distribution $\bar{\pi}$, the long-run fraction of time each cell should be occupied, and we must compute the multipliers λ_c that makes the kernel realise it. For uniform coverage we set $\bar{\pi}_k = 1/K$, so every cell is equally visited at equilibrium regardless of how many neighbours it has. This problem is non-trivial precisely because the graph is irregular. On a regular graph, equal multipliers give a uniform stationary distribution and there is nothing to solve. On the irregular 8-connected 20×20 grid, equal multipliers give a degree-weighted stationary distribution, so cells with more neighbours are visited more often. Correcting for this requires the multiplier on each cell to depend on its position in the graph, which is what the inverse solver computes.

A.2 A closed-form expression for the stationary distribution. The first step is to derive the stationary distribution as an explicit function of the multipliers, which makes the inverse problem a system of equations rather than a black box. Introduce the change of variable $b_k = \exp(-\lambda_c^k)$. Since the exponential is positive, $b_k > 0$ always. Substituting into the Layer 1-C kernel from Sec. 3.1:

$$p^*(k_2|k_1) = \frac{A_{\{k_1 k_2\}} b_{\{k_2\}}}{\sum_j A_{\{k_1 j\}} b_j} = \frac{A_{\{k_1 k_2\}} b_{\{k_2\}}}{(Ab)_{\{k_1\}}}$$

where $(Ab)_k = \sum_j A_{\{k j\}} b_j$ is the sum of b over the neighbours of cell k . The kernel is now linear in b in both numerator and denominator. Because the underlying graph is undirected, the kernel is reversible: it satisfies detailed balance with respect to its own stationary distribution. Detailed balance says

$$\pi_{\{k_1\}} p^*(k_1|k_2) = \pi_{\{k_2\}} p^*(k_2|k_1)$$

for every adjacent pair (k_1, k_2) . Substituting the kernel and using $A_{\{k_1 k_2\}} = A_{\{k_2 k_1\}}$ (symmetry of the adjacency matrix) gives, after cancellation,

$$\frac{\pi_{\{k_1\}}}{[b_{\{k_1\}}(Ab)_{\{k_1\}}]} = \frac{\pi_{\{k_2\}}}{[b_{\{k_2\}}(Ab)_{\{k_2\}}]}$$

This ratio is constant across every edge of a connected graph, hence constant for all k .

Combining with $\sum_k \pi_k = 1$ gives the closed form

$$\pi_k(b) = \frac{b_k (Ab)_k}{\sum_j b_j (Ab)_j} \quad (\text{A.1})$$

The denominator is a normalisation scalar (it equals $b^T A b$, a single number). The inverse problem now has a clean statement: find a positive vector b such that the right-hand side of (A.1) equals the prescribed target $\bar{\pi}$. Equivalently, since the denominator is just a normaliser, find b such that

$$b_k (Ab)_k \propto \bar{\pi}_k, \forall k \quad (\text{A.2})$$

A.3 Recognising symmetric matrix scaling.

Equation (A.2) is a coupled system of K nonlinear equations in K unknowns. There is no closed-form solution on an irregular graph, because $(Ab)_k$ depends on the values of b at the neighbours of cell k , which in turn depend on the values at their neighbours, and so on. What it does match is a classical problem from linear algebra: symmetric matrix scaling. Given a non-negative matrix A and a target marginal $\bar{\pi}$, scaling theory asks whether one can find a positive diagonal matrix $D = \text{diag}(b)$ such that the rescaled matrix DAD has prescribed row (and column) sums proportional to the target distribution. The row sum of DAD at row k is exactly $b_k (Ab)_k$, so (A.2) is precisely the row-sum condition of the symmetric scaling problem.

This problem has been studied since Sinkhorn's 1964 paper [12], with the symmetric version covered explicitly in Sinkhorn and Knopp [13] and reviewed comprehensively by Idel [14, Sec.

5]. The classical result (Sinkhorn's theorem) guarantees that a solution exists and is unique up to a multiplicative constant on b whenever A is non-negative and fully indecomposable. For undirected graphs, full indecomposability is equivalent to the graph being connected and having at least one self-loop or odd cycle, both of which our 8-connected 20×20 grid satisfies trivially. Sinkhorn's result is purely an existence-and-uniqueness statement; it does not by itself give us b . For that we need an algorithm.

A.4 The symmetric IPF iteration. The standard iterative algorithm for matrix scaling is the Iterative Proportional Fitting Procedure (IPF), also known as the Sinkhorn–Knopp algorithm or the RAS method. The asymmetric version alternates two passes: rescale all rows so their sums match a target, then rescale all columns so their sums match a target, then rescale rows again, and so on, until both are simultaneously matched. This is the algorithm of [13]; see [14, Sec. 4] for a modern exposition.

In our setting the row and column scalings would have to be the same vector b , because the matrix DAD uses the same D on both sides. The asymmetric iteration's two passes therefore collapse into a single symmetric step. Idel [14, Algorithm 5.9] gives the symmetric form: at iteration m , evaluate the current stationary distribution $\pi^m = \pi(b^m)$ from (A.1), then update

$$b_k^{(m+1)} = b_k^m \cdot \sqrt{\pi_k / \pi_k^m} \quad (\text{A.3})$$

Three things deserve unpacking.

Why a multiplicative update: the constraint (A.2) is multiplicative in b_k , so the natural correction is to multiply b_k by the ratio between where we want it to be and where it currently is, the same logic that drives the asymmetric Sinkhorn–Knopp iteration.

Why the square root: the asymmetric iteration multiplies by the full ratio $\frac{\pi_k}{\pi_k^m}$ in the row pass and again in the column pass; in the symmetric setting both passes act on the same vector entry b_k , so applying the full ratio in a single step would over-correct by roughly a factor of two. The square root is the geometric-mean compromise, half the correction applied

symmetrically, and is what makes the iteration stable. This is derived rigorously in [14, Sec. 5] and was studied earlier by Knight (2008) for the symmetric scaling problem specifically.

Why it converges: the classical convergence theorem of [13] states that for non-negative, fully indecomposable A and strictly positive target marginals, the iteration converges geometrically, the residual at iteration m shrinks by a constant factor each step. For the symmetric variant, [14, Thm. 5.10] gives the convergence rate explicitly. On our grid, "geometrically" turns out to mean that 89 iterations are enough to drive the residual below double-precision noise, which matches the predicted rate.

A.5 Gauge fixing. The kernel of Sec. A.2 has an exact symmetry: replacing b by $c \cdot b$ for any positive constant c leaves both numerator and denominator of $p^*(k_2 | k_1)$ scaled by the same factor c , so the kernel is unchanged. This means a whole one-parameter family of vectors $\{c \cdot b : c > 0\}$ produces identical swarm behaviour. We say b is determined only up to a multiplicative constant. Equivalently, since $\lambda_c^k = -\log b_k$, replacing b by $c \cdot b$ adds $-\log c$ to every multiplier; we say λ_c is determined only up to an additive constant. Only differences $\lambda_c^{k_1} - \lambda_c^{k_2}$ are physically meaningful; absolute values are not.

This freedom is benign at the level of the final answer (any solution we find is just as valid as any other up to the constant), but it causes a numerical problem during the iteration: nothing in (A.3) prevents the iterates from drifting uniformly upward or downward in scale. Even if the ratios b_k / b_j are converging beautifully, the absolute values can grow or shrink unboundedly, eventually causing numerical overflow or underflow. We therefore fix the gauge after each update by normalising the geometric mean of b to one:

$$b_k^{(m+1)} \leftarrow \frac{b_k^{(m+1)}}{\exp[\frac{1}{K} \sum_j \log b_j^{(m+1)}]} \quad (\text{A.4})$$

This rescaling does not change the kernel (by the symmetry above), only the numerical scale of b . After taking logs, (A.4) becomes the equivalent

statement $\sum_k \lambda_c^k = 0$: the multipliers are centred around zero. This is the natural choice for a symmetric problem where there is no privileged "reference cell" to anchor to, and it is the convention adopted in our implementation.

A.6 Stopping criterion and recovery. We declare convergence when the maximum per-component change drops below tolerance: $\|b^{(m+1)} - b^m\|_{\inf} = \max_k |b_k^{(m+1)} - b_k^{(m)}| < 10^{-11}$.

The infinity-norm is stricter than the L^1 or L^2 alternatives: it requires every component to be near-stationary, not just the average. This is appropriate here because each cell's multiplier matters individually for downstream behaviour. Once b has converged, the multipliers are recovered by inverting the change of variable: $\lambda_c^k = -\log b_k$, followed by a final mean-zero shift (equivalent to the gauge fixing of Sec. A.5 applied to λ_c directly). This is the multiplier field reported in Sec. 4.3.2 and used in the hierarchical controller.

The implementation validates the result two ways. The theoretical check substitutes b back into (A.1) and reports $\|\pi(b) - \bar{\pi}\|_1$, which captures how well the iteration solved the equations; the archived value 3.32×10^{-11} is the residual at the convergence floor. The empirical check runs the actual simulator with the recovered λ_c , accumulates Markov arrivals, and reports the empirical $\|c - \bar{\pi}\|_1$; the archived value 4.29×10^{-2} is finite-sample fluctuation around the designed target. The 9-order-of-magnitude gap between the two is sampling noise, not a deficiency of the solver.

Appendix B — Information Diffusion: Fixed-Point and Inverse-Rate Derivations

B.1 The self-consistent fixed-point equation.

The Layer 1-I kernel of Sec. 3.1 has a structure that the coverage kernel does not: the stationary distribution $\bar{\pi}$ appears inside the kernel that produces it. To make this explicit, define

$c_k(\pi) = \lambda_c^k + \lambda_I \pi_k$ and $b_k(\pi) = \exp(-c_k(\pi))$, so the kernel reads $p^*(k_2 | k_1) \propto A_{k_1 k_2} b_{k_2}(\pi)$. For

any candidate π , the stationary distribution of the kernel built from π is given by the same closed form as in Appendix A, namely

$$F(\pi)_k = \frac{b_k(\pi) (Ab(\pi))_k}{\sum_j b_j(\pi) (Ab(\pi))_j} \quad (\text{B.1})$$

A self-consistent solution is a $\bar{\pi}$ that, when fed into the kernel as a parameter, produces itself as the stationary distribution. This is the fixed-point equation

$$\bar{\pi} = F(\bar{\pi}) \quad (\text{B.2})$$

Brouwer's fixed-point theorem (which Pinciroli invokes in [5]) guarantees that a solution exists, because F is a continuous map from the probability simplex to itself. It does not guarantee uniqueness, and for $\lambda_I < 0$ it is genuinely non-unique: there are typically a coverage-like fixed point near the degree-weighted distribution, and one or more clustered fixed points where the swarm concentrates on a few cells. Picking among them is part of the design problem.

B.2 Why this is fundamentally different from the coverage inverse. Both Appendix A and Appendix B solve self-referential equations involving the stationary distribution, but the unknowns are different. In the coverage inverse, $\bar{\pi}$ is given and we solve for the multipliers; the equations have rich algebraic structure inherited from matrix scaling, and an iterative scaling procedure converges geometrically. In the information-diffusion problem, the multiplier λ_I is given (a single scalar) and we solve for $\bar{\pi}$, an entire vector. There is no marginal-matching structure to exploit, the problem is just a generic fixed-point equation in R^K . We need a different algorithm.

B.3 Why plain iteration fails: contraction and non-contraction.

The most natural way to solve $\bar{\pi} = F(\bar{\pi})$ is to iterate: pick any starting π^0 and compute $\pi^{(m+1)} = F(\pi^m)$. This is plain Picard iteration, also called fixed-point iteration. Whether it converges depends on a single property of F .

A map F is called a contraction with factor $L < 1$ if, for every pair of inputs x, y ,

$$\|F(x) - F(y)\| \leq L \cdot \|x - y\|$$

Distances shrink by at least a factor L on every application. The Banach fixed-point theorem then guarantees that Picard iteration converges to the unique fixed point of F from any starting point, with the residual at iteration m bounded by L^m times the initial residual. A contraction is a self-correcting map: any deviation gets damped out.

A map that is not a contraction can do anything. It might still have a fixed point, but iterating F doesn't necessarily find it. The iterates can oscillate, settle into a limit cycle, drift away, or jump erratically between basins of attraction. For the Layer 1-I map, this is exactly what happens when $\lambda_l < 0$. The fixed-point map F encodes positive feedback: high-density cells attract more robots ($\lambda_l < 0$ rewards them), which raises their density further, which attracts even more robots. Two slightly-different candidate stationary distributions π and π' get pushed further apart by F rather than closer together, because each one's high-density cells get amplified differently. F is not a contraction in this regime, in technical terms, its Jacobian has eigenvalues larger than one in magnitude. Plain Picard iteration on such a map can produce oscillating sequences that never settle, even when the fixed point itself is well-defined.

B.4 The damped (Krasnosel'skii-Mann)

iteration. The standard remedy for non-contractive Picard iteration is the relaxed or damped form

$$\pi^{(m+1)} = (1 - \alpha) * \pi^{(m)} + \alpha F(\pi^{(m)}), \alpha \in (0, 1) \quad (\text{B.3})$$

Each step takes only a fraction α of the way toward the naive Picard step $F(\pi^{(m)})$, retaining a fraction $(1 - \alpha)$ of the previous iterate. This is known as a Krasnosel'skii-Mann iteration or a Mann iteration, and is treated comprehensively in Bauschke and Combettes [15, Ch. 5]; the relaxed-Picard form for fixed-point problems is also covered classically in Ortega and Rheinboldt [16, Ch. 10].

The key theoretical property is this. Even if F is not a contraction, as long as F is non-expansive, meaning $\|F(x) - F(y)\| \leq \|x - y\|$ (distances do not grow), the relaxed iteration (B.3) is guaranteed to converge to a fixed point of F from any starting point [15, Thm. 5.15].

Non-expansiveness is a much weaker condition than contractivity, and the Layer 1-I map satisfies it on bounded subsets of the simplex.

Two important properties to keep in mind. First, damping does not change what equation we are solving. Any π satisfying $\pi = F(\pi)$ also satisfies $\pi = (1 - \alpha)\pi + \alpha F(\pi)$, by simple algebra. The relaxed iteration is therefore looking for the same fixed points as plain Picard; it just takes a different path to find them. Second, damping does not select among multiple fixed points. If (B.2) has several solutions, the relaxed iteration will converge to one of them, and which one depends on the starting point. We will exploit this dependence in B.5. The implementation uses $\alpha = 0.5$, which is the conventional default in the Mann-iteration literature [15, Sec. 5.2]: each step is the geometric midpoint between the current iterate and the naive Picard step. Smaller values give more conservative, slower iterations; larger values approach plain Picard and risk instability. We declare convergence when

$\|\pi^{(m+1)} - \pi^{(m)}\|_{inf} < 10^{-11}$, i.e., the maximum per-cell change between successive iterates falls below tolerance. As in Appendix A, the infinity-norm is the strictest of the standard choices and is appropriate when each cell's value matters individually, which it does here, because

the encounter probability $p_{enc} = \sum_k^K \pi_k^2$ is sensitive to outliers.

B.5 Branch-aware continuation: choosing among multiple fixed points.

For $\lambda_l < 0$, equation (B.2) has multiple solutions, and the design problem requires us to specify which one we are following. The relaxed iteration of B.4 converges to the fixed point closest to its starting point, so we use the starting point as the branch selector.

Specifically, we trace each branch by continuation in λ_l , a standard technique from numerical analysis for following solution

branches of parameter-dependent nonlinear equations [16, Ch. 10]. The idea is that the fixed point depends smoothly on λ_I (within a single branch), so a small change in λ_I produces a small change in the fixed point. If we already have the solution $\bar{\pi}(\lambda_I)$, we can find the solution at a nearby value $\lambda_I + \Delta\lambda_I$ by re-running the relaxed iteration with $\bar{\pi}(\lambda_I)$ as the starting point. The new iteration converges to the nearby fixed point on the same branch, not to a possibly very different fixed point on a different branch, because the starting point places us inside the new branch's basin of attraction.

The two branches of interest are traced as follows. The coverage-like branch is initialised at $\lambda_I = 0$, where the fixed point is unique and equals the degree-weighted distribution with coverage multipliers set to a constant. We then decrease λ_I in small steps, warm-starting each iteration from the previous fixed point. The fixed point varies smoothly and stays close to the coverage-like distribution; the encounter probability p_{enc} stays close to its zero-multiplier baseline. The clustered branch is initialised at a strongly negative λ_I (e.g., $\lambda_I = -200$), where the clustered fixed point is well-separated from any coverage-like solution and is reliably reached by the iteration when started from a perturbed concentrated distribution. We then continue upward in λ_I to map out the branch's extent, again warm-starting each iteration from the previous fixed point. On our grid, this branch turns out to be highly degenerate: across a wide range of strongly negative λ_I the fixed point is similar (concentrated on a small number of cells) and the encounter probability p_{enc} saturates near 0.5.

B.6 The outer scalar inverse: matching a target spreading rate.

The fixed-point solve of B.4 and the branch tracing of B.5 together solve the forward Layer 1-I problem: given λ_I and a branch choice, return the fixed-point stationary distribution. The actual inverse Layer 1-I problem is one level higher: given a target spreading rate, solve for the scalar λ_I .

The connection is the mean-field model from Sec. 3.1. The spreading rate is

$\frac{dn}{dt} = A \beta p_{enc} (n/A) (1 - (n/A))$, where A is the swarm size, β is the per-encounter transmission probability, $f = n/A$ is the informed fraction, and $p_{enc} = \sum_k \pi_k^2$ is the cell-coincidence probability computed from the stationary field. Given a target rate $r^* = dn/dt$, we convert it into a target encounter probability

$$p_{enc, target} = r^* / [A \beta f(1 - f)], \quad (B.4)$$

and the inverse problem becomes: find a λ_I on the chosen branch such that

$$p_{enc}(\bar{\pi}(\lambda_I)) = p_{enc, target}.$$

This is a one-dimensional root-finding problem in λ_I . The complication is that each evaluation of $p_{enc}(\lambda_I)$ requires running the inner fixed-point solve of B.4 to convergence, which is moderately expensive. We therefore want a method that minimises the number of function evaluations while remaining robust against an objective function that is potentially noisy (because the inner solve has finite tolerance) and has saturated plateaus (the clustered branch). We use a continuation-then-bisection strategy, which combines two standard techniques. Continuation trace: first, we sweep λ_I across a range of values along the chosen branch (using the warm-starting strategy of B.5), evaluating $p_{enc}(\lambda_I)$ at each. This produces a tabulated curve of p_{enc} versus λ_I , and we walk along it looking for a bracketing pair: two consecutive trace values $\lambda_I^{(j)}$ and $\lambda_I^{(j+1)}$ such that $p_{enc, target}$ lies between $p_{enc}(\lambda_I^{(j)})$ and $p_{enc}(\lambda_I^{(j+1)})$. By the intermediate value theorem, the target is achieved somewhere between them.

Bisection: given a bracketing pair, we refine by bisection, a textbook root-finding method that halves the bracket on each step. Specifically, evaluate p_{enc} at the midpoint

$\lambda_I^{mid} = (\lambda_I^{(j)} + \lambda_I^{(j+1)})/2$; whichever half of the bracket still contains the target becomes the new bracket; repeat. After n steps the bracket has

length $\frac{|\lambda_l^{(j+1)} - \lambda_l^{(j)}|}{2}$, so convergence is geometric (each step halves the uncertainty). Bisection requires only function evaluations and a sign change to work, with no assumptions about smoothness or differentiability.

We use bisection rather than faster alternatives, Newton's method, secant method, regula falsi, for two reasons. First, the inner fixed-point solve provides $p_{enc}(\lambda_l)$ as a noisy point evaluation, with no derivative information. Methods that need a derivative would require finite differences, which compound the inner-solve noise; methods that approximate a derivative from successive evaluations are sensitive to that noise. Second, bisection is unconditionally robust: as long as the bracket genuinely contains a root, bisection cannot fail, regardless of how badly behaved the function is. For an expensive, noisy inner objective this robustness matters more than asymptotic speed.

The "weakest-magnitude" rule: when the target value is achieved by multiple λ_l on the same branch, which happens here because the clustered branch saturates at $p_{enc} \approx 0.5$ for a wide range of λ_l , the solver returns the λ_l with smallest $|\lambda_l|$ that still satisfies the tolerance. This implements a principle of least intervention: if multiple multipliers all achieve the goal, choose the smallest one. The behaviour produced by $\lambda_l = -20$ on the saturated plateau is essentially indistinguishable from that produced by $\lambda_l = -200$, so reporting -20 is both numerically and philosophically the right answer: it is the smallest bias the controller has to apply.

Failure mode: when the target $p_{enc, target}$ falls outside the reachable range on the chosen branch, for example, asking for $p_{enc, target} = 0.8$ when the branch saturates at 0.5 , the continuation trace finds no bracketing pair and the solver reports non-convergence. This is the correct behaviour: the target is genuinely infeasible for this controller on this graph, and silently returning a "best effort" value would be misleading.